

- 2 Lichens consist of protocists and fungi living in close association. Both types of organism benefit from this association. Lichens are found in a large range of habitats, including on the trunks of trees.

Fig. 2.1 shows an example of a lichen, the salted shield lichen, *Parmelia saxatilis*, on the trunk of a tree.

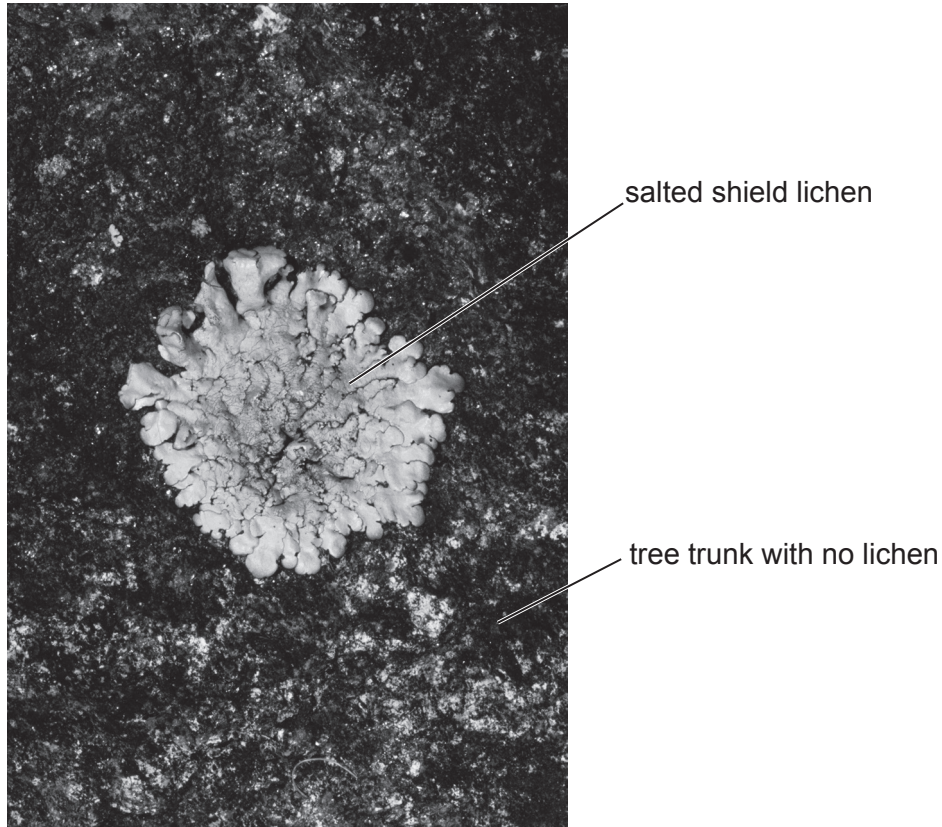


Fig. 2.1

The red deer, *Cervus elaphus*, is found in many parts of the world including north-west Europe. Red deer are found in a large range of habitats, including mixed woodlands. Mixed woodlands contain many different tree species.

Fig. 2.2 shows a male red deer.



Fig. 2.2

Red deer have increased in numbers in the mountainous regions of Scotland in north-west Europe since the 1960s.

Scientists think that grazing of plants by red deer has affected the abundance of certain lichens found growing on the trunks of trees in mixed woodlands.

To investigate this, the scientists excluded red deer from certain mixed woodland areas (exclosures) in 1995, allowing the exclosures to remain ungrazed.

In 2013, the scientists compared the abundance of the salted shield lichen in the exclosures and in the grazed areas.

The scientists used the grid shown in Fig. 2.3 to determine the abundance of the salted shield lichen on the tree trunks.

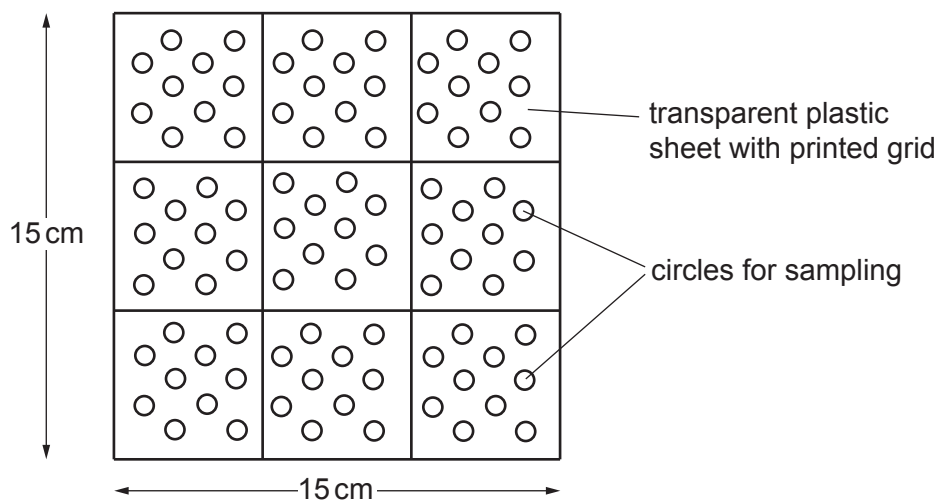


Fig. 2.3

For each tree sampled:

- The grid was placed against the tree trunk at three different heights: base, middle and upper.
- The scientists counted the number of circles that contained the salted shield lichen.
- The number of circles that contained the salted shield lichen was used to calculate the percentage cover of salted shield lichen on the tree.

This study was carried out over the course of the three months (June, July and August) in the summer of 2013 in the exclosures and the grazed areas.

(a) (i) Suggest **two** variables that the scientists should have standardised in this investigation.

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..... [2]

(ii) State **one** risk **and** the safety precaution that the scientists should take when measuring the abundance of salted shield lichen on the tree trunks.

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..... [2]

(b) The scientists calculated a percentage cover for the salted shield lichen as **63.3%** on one of the trees, using the grid in Fig. 2.3.

Calculate how many circles on the grid contained salted shield lichen.

Show your working and give your answer to the nearest whole number.

number of circles with salted shield lichen = [2]

- (c) The scientists performed a t -test to compare the abundance of the salted shield lichen in the enclosures and the grazed areas.

- (i) State a null hypothesis for the t -test.

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 [1]

Table 2.1 shows the mean results and values of t the scientists obtained.

Table 2.1

position of grid on tree trunk	mean percentage cover in enclosures	mean percentage cover in grazed areas	t	significance at $p = 0.05$
base	1.7	3.2	0.730	not significant
middle	3.4	9.8	2.117	significant
upper	8.5	22.0	2.071	significant

- (ii) State **one** conclusion that can be made from the data in Table 2.1.

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 [1]

- (iii) State **two** improvements the scientists could make to their study to determine the effect of grazing of red deer on the abundance of salted shield lichen in mixed woodland areas.

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 [2]

[Total: 10]

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